Section 1: Mechanics I — Solutions

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1. Projectile Motion

Given:

$$v_0 = 100 \text{ m/s}, \qquad \theta = 37^\circ, \qquad g = 9.81 \text{ m/s}^2$$

Resolve the initial velocity into components:

$$v_{0x} = v_0 \cos\theta, \qquad v_{0y} = v_0 \sin\theta$$

Differential equations of motion

In the horizontal direction there is no acceleration:

$$\frac{d^2x}{dt^2} = 0$$

In the vertical direction gravity acts downward:

$$\frac{d^2y}{dt^2} = -g$$

With initial conditions

$$x(0) = 0, \quad y(0) = 0, \quad \frac{dx}{dt}(0) = v_0 \cos\theta, \quad \frac{dy}{dt}(0) = v_0 \sin\theta$$

Integrating gives:

$$x(t) = v_0 \cos\theta \, t$$

$$y(t) = v_0 \sin\theta \, t - \frac{1}{2} g t^2$$

Time of flight

The projectile lands when $y(T) = 0$ with $T > 0$:

$$v_0 \sin\theta \, T - \frac{1}{2} g T^2 = 0$$

$$T \left(v_0 \sin\theta - \frac{gT}{2} \right) = 0$$

Hence,

$$T = \frac{2 v_0 \sin\theta}{g}$$

Numerically,

$$T = \frac{2(100)\sin 37^\circ}{9.81} \approx 12.27 \text{ s}$$

Maximum height

At the top of the trajectory, the vertical velocity is zero:

$$v_y(t) = v_0 \sin\theta - gt$$

So the time to reach maximum height is

$$t_{\max} = \frac{v_0 \sin\theta}{g}$$

Substitute into $y(t)$:

$$H_{\max} = \frac{v_0^2 \sin^2\theta}{2g}$$

Numerically,

$$H_{\max} = \frac{100^2 \sin^2 37^\circ}{2(9.81)} \approx 184.6 \text{ m}$$

Range

The horizontal range is

$$R = x(T) = v_0 \cos\theta \cdot T$$

$$R = v_0 \cos\theta \cdot \frac{2v_0 \sin\theta}{g} = \frac{v_0^2 \sin(2\theta)}{g}$$

Numerically,

$$R = \frac{100^2 \sin 74^\circ}{9.81} \approx 979.9 \text{ m}$$

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2. Range Optimization

For projectile motion,

$$R(\theta) = \frac{v_0^2 \sin(2\theta)}{g}$$

Since v_0^2/g is constant, maximizing $R(\theta)$ is equivalent to maximizing $\sin(2\theta)$.

Differentiate:

$$\frac{dR}{d\theta} = \frac{v_0^2}{g} \cdot 2\cos(2\theta)$$

Set the derivative to zero:

$$\cos(2\theta) = 0$$

$$2\theta = \frac{\pi}{2} \quad \Rightarrow \quad \theta = \frac{\pi}{4} = 45^\circ$$

Also, $\sin(2\theta)$ reaches its maximum possible value, 1, when $2\theta = \pi/2$.

Therefore the maximum range is achieved at

$$\boxed{\theta = 45^\circ}$$

and the maximum range itself is

$$R_{\max} = \frac{v_0^2}{g}$$

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3. Path Intersection

Alice:

$$A(t) = (2 + t,\ 8 - 3t)$$

Bob:

$$B(t) = (2t - 1,\ 2t + 2)$$

Do they collide at the same time?

For a collision we need $A(t) = B(t)$ for the same t.

From the x-coordinates:

$$2 + t = 2t - 1 \quad \Rightarrow \quad t = 3$$

Check the y-coordinates at $t = 3$:

$$8 - 3(3) = -1, \qquad 2(3) + 2 = 8$$

These are not equal, so they do not collide.

Do the geometric paths intersect?

Allow different parameters s and u:

$$(2+s,\ 8-3s) = (2u-1,\ 2u+2)$$

From the first equation,

$$u = \frac{s+3}{2}$$

Substitute into the second:

$$8-3s = 2\left(\frac{s+3}{2}\right) + 2 = s+5$$

$$3 = 4s \quad \Rightarrow \quad s = \frac{3}{4}$$

Then

$$u = \frac{\frac{3}{4}+3}{2} = \frac{15}{8}$$

The paths intersect at the point

$$\left(2+\frac{3}{4},\ 8-3\cdot\frac{3}{4}\right) = \left(\frac{11}{4},\ \frac{23}{4}\right)$$

So the paths intersect geometrically, but the two people are not there at the same time.

Minimum distance between them

At equal time t, the separation vector is

$$A(t) - B(t) = \big((2+t)-(2t-1),\ (8-3t)-(2t+2)\big) = (3-t,\ 6-5t)$$

Hence

$$D^2(t) = (3-t)^2 + (6-5t)^2$$

$$D^2(t) = 26t^2 - 66t + 45$$

Minimize this quadratic:

$$\frac{d}{dt}D^2(t) = 52t - 66 = 0$$

$$t = \frac{33}{26} \approx 1.269 \text{ s}$$

The minimum squared distance is

$$D_{\min}^2 = 26\left(\frac{33}{26}\right)^2 - 66\left(\frac{33}{26}\right) + 45 = \frac{81}{26}$$

Therefore,

$$D_{\min} = \frac{9}{\sqrt{26}} \approx 1.77$$

Conclusion:

$$\boxed{\text{No collision occurs. The minimum distance is } \frac{9}{\sqrt{26}} \text{ at } t = \frac{33}{26}.}$$

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4. Vector Calculus

Given

$$\vec{r}(t) = (3t^2)\hat{\imath} + (5t - 8t^2)\hat{\jmath}$$

Velocity is the derivative of position:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = (6t)\hat{\imath} + (5 - 16t)\hat{\jmath}$$

Acceleration is the derivative of velocity:

$$\vec{a}(t) = \frac{d\vec{v}}{dt} = 6\hat{\imath} - 16\hat{\jmath}$$

So,

$$\boxed{\vec{v}(t) = 6t\hat{\imath} + (5 - 16t)\hat{\jmath}}$$

$$\boxed{\vec{a}(t) = 6\hat{\imath} - 16\hat{\jmath}}$$

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5. Relative Velocity

River velocity relative to the ground:

$$\vec{v}_r = 2\hat{\imath}$$

Let the boat head at an angle α west of north. Its velocity relative to the water is

$$\vec{v}_{bw} = -5\sin\alpha\,\hat{\imath} + 5\cos\alpha\,\hat{\jmath}$$

The boat wants to move directly north relative to the ground, so the total east-west component must be zero:

$$-5\sin\alpha + 2 = 0$$

$$\sin\alpha = \frac{2}{5}$$

$$\alpha = \sin^{-1}\left(\frac{2}{5}\right) \approx 23.6^\circ$$

So the boat should head

$$\boxed{23.6^\circ \text{ west of north}}$$

The northward speed across the river is then

$$v_{\text{north}} = 5\cos\alpha = 5\sqrt{1 - \left(\frac{2}{5}\right)^2} = \sqrt{21} \approx 4.58 \text{ m/s}$$

If the river is 200 m wide, the crossing time is

$$t = \frac{200}{\sqrt{21}} \approx 43.6 \text{ s}$$

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6. Variable Velocity

Given

$$v(t) = t^2 + 2t - 5$$

Acceleration is

$$a(t) = \frac{dv}{dt} = 2t + 2$$

Position is the integral of velocity:

$$x(t) = \int (t^2 + 2t - 5)\,dt = \frac{t^3}{3} + t^2 - 5t + C$$

Use $x(0) = 4$:

$$4 = C$$

So

$$x(t) = \frac{t^3}{3} + t^2 - 5t + 4$$

At $t = 3$,

$$x(3) = \frac{27}{3} + 9 - 15 + 4 = 7$$

$$a(3) = 2(3) + 2 = 8$$

Therefore,

$$\boxed{x(3) = 7}$$

$$\boxed{a(3) = 8}$$

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7. Elimination of Time and Interpretation of Acceleration

Given

$$x(t) = 2t^2, \qquad y(t) = 3t^3$$

Eliminate the parameter

From $x = 2t^2$,

$$t^2 = \frac{x}{2}$$

Square $y = 3t^3$ to avoid the sign ambiguity:

$$y^2 = 9t^6 = 9\left(\frac{x}{2}\right)^3 = \frac{9}{8}x^3$$

So the Cartesian equation is

$$\boxed{y^2 = \frac{9}{8}x^3, \qquad x \geq 0}$$

This is a semicubical parabola. For $t > 0$, the curve lies above the x-axis; for $t < 0$, it lies below it.

Trajectory sketch

The trajectory starts at the origin and opens to the right, with two branches symmetric about the x-axis.

Velocity and speed

$$\vec{v}(t) = \left(\frac{dx}{dt}, \frac{dy}{dt}\right) = (4t,\ 9t^2)$$

$$|\vec{v}(t)| = \sqrt{(4t)^2 + (9t^2)^2} = \sqrt{16t^2 + 81t^4} = |t|\sqrt{16 + 81t^2}$$

Acceleration and its magnitude

$$\vec{a}(t) = \left(\frac{d^2x}{dt^2}, \frac{d^2y}{dt^2}\right) = (4,\ 18t)$$

$$|\vec{a}(t)| = \sqrt{4^2 + (18t)^2} = \sqrt{16 + 324t^2}$$

Is the acceleration constant?

No. The x-component is constant, but the y-component is $18t$, which depends on time. Therefore,

$$\boxed{\vec{a}(t) \text{ is not constant}}$$

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8. Circular Motion

At the equator, the centripetal acceleration due to Earth's rotation is

$$a_c = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

Take

$$R = 6378 \text{ km} = 6.378 \times 10^6 \text{ m}, \qquad T = 24 \times 3600 = 86400 \text{ s}$$

Then

$$a_c = \frac{4\pi^2(6.378 \times 10^6)}{86400^2} \approx 3.37 \times 10^{-2} \text{ m/s}^2$$

So,

$$\boxed{a_c \approx 0.0337 \text{ m/s}^2}$$

This is about 0.34% of g.

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9. Momentum Comparison

Momentum is

$$p = mv$$

Fly

$$m_f = 2 \text{ g} = 0.002 \text{ kg}, \qquad v_f = 10 \text{ m/s}$$

$$p_f = 0.002 \cdot 10 = 0.020 \text{ kg m/s}$$

Tennis ball

$$m_t = 60 \text{ g} = 0.060 \text{ kg}, \qquad v_t = 1 \text{ m/s}$$

$$p_t = 0.060 \cdot 1 = 0.060 \text{ kg m/s}$$

Since

$$0.060 > 0.020$$

the tennis ball has the greater momentum.

$$\boxed{\text{The tennis ball has 3 times the momentum of the fly.}}$$

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10. Kinematics

Given

$$\vec{r}(t) = \big(a\cos(\omega t),\ b\sin(\omega t),\ bt\big)$$

a) Equation of the trajectory

From the first two coordinates,

$$x = a\cos(\omega t), \qquad y = b\sin(\omega t)$$

Hence,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \cos^2(\omega t) + \sin^2(\omega t) = 1$$

The projection onto the xy-plane is therefore an ellipse.

Because

$$z = bt$$

the point rises linearly in z, so the full 3D trajectory is an elliptical helix:

$$\boxed{\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad z = bt}$$

Equivalently, eliminating t in favor of z gives

$$x = a\cos\left(\frac{\omega z}{b}\right), \qquad y = b\sin\left(\frac{\omega z}{b}\right)$$

b) Path length from $t = 0$ to $t = t_0$

Differentiate:

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = (-a\omega \sin(\omega t),\ b\omega \cos(\omega t),\ b)$$

Its magnitude is

$$|\vec{v}(t)| = \sqrt{a^2\omega^2 \sin^2(\omega t) + b^2\omega^2 \cos^2(\omega t) + b^2}$$

Therefore the arc length is

$$\boxed{s(t_0) = \int_0^{t_0} \sqrt{a^2\omega^2 \sin^2(\omega t) + b^2\omega^2 \cos^2(\omega t) + b^2}\, dt}$$

In general this integral is expressed in terms of elliptic integrals.

Special case $a = b = R$:

$$|\vec{v}(t)| = \sqrt{R^2\omega^2 + b^2}$$

which is constant, so

$$s(t_0) = t_0\sqrt{R^2\omega^2 + b^2}$$

c) Python visualization and special cases

Example Python code:

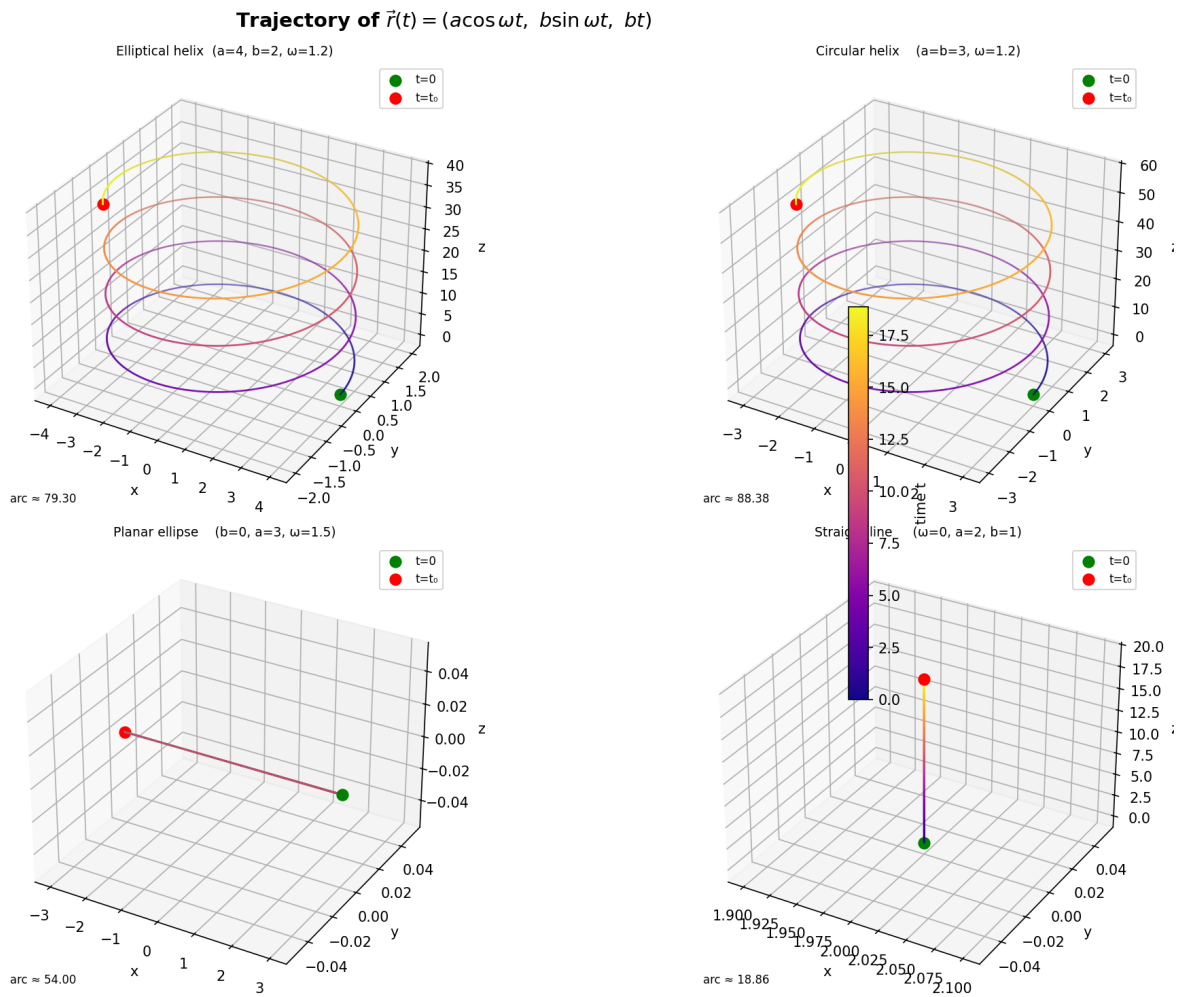
```
import numpy as np
import matplotlib.pyplot as plt

a = 4
b = 2
omega = 1.2
t0 = 8 * np.pi

t = np.linspace(0, t0, 1200)
x = a * np.cos(omega * t)
y = b * np.sin(omega * t)
z = b * t

fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection="3d")
ax.plot(x, y, z, linewidth=2)

ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_zlabel("z")
ax.set_title("Elliptical Helix")
plt.show()
```

Special cases:

- If $a = b$, the projection in the xy-plane is a circle, so the trajectory becomes a circular helix.
- If $\omega = 0$, then $x = a$, $y = 0$, and the motion is a straight vertical line.
- If $b = 0$, then $y = 0$ and $z = 0$, so the motion collapses to oscillation along the x-axis.